Coinciding transition attractors need not attract at the threshold

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Task 1. Resolve MathDB 354469. Conjecture 6.1 of Leśniak, Snigireva, Strobin and Vince [1], p. 5425, asks whether, under (H1)–(H3), equality of a lower transition attractor with one upper transition attractor forces uniqueness of the upper transition attractor and attraction by the terminal IFS. The terminal-attraction assertion is false.

Here (H1) requires continuous dependence of each map on the parameter at every point; (H2) requires a Lipschitz constant below one for each parameter $t < 1$; and (H3) requires the individual fixed points to have limits as $t \uparrow 1$. The source defines an attractor using Hausdorff convergence from every nonempty compact subset of some open neighborhood (p. 5398). Mere invariance is insufficient.

Proposition 1. *On the complete metric space $\mathbb{R}$ with its usual distance, put*

$$r_t = \frac{1+t}{2}, \qquad F_t = \{g_t, f_t\}, \qquad g_t(x) = r_t x, \quad f_t(x) = \frac{r_t}{2}x \quad (0 \le t \le 1).$$

This family satisfies (H1)–(H3), has $A_\bullet = A^\bullet = \{0\}$ and a unique upper transition attractor, but $\{0\}$ is not an attractor of F_1.

Proof. The two maps are distinct, invertible and continuously parameterized, including at $t = 0$. For $0 \le t < 1$, their maximal Lipschitz constant is $r_t < 1$, and both have the unique fixed point zero. Thus (H1)–(H3) hold and the set Q of limiting fixed points is $\{0\}$.

For every nonempty compact $K \subset \mathbb{R}$ and integer $n \ge 0$,

$$F_t^n(K) = \bigcup_{k=0}^{n} r_t^n 2^{-k} K. \tag{1}$$

Indeed, a word of length n containing k copies of f_t multiplies every point by $r_t^n 2^{-k}$, and every k occurs. Writing $M_K = \max_{x\in K} |x|$, equation (1) gives the exact formula

$$h(F_t^n(K), \{0\}) = r_t^n M_K. \tag{2}$$

Consequently $A_t = \{0\}$ is the attractor for every $t < 1$, with basin $\mathbb{R}$. In particular every increasing sequence of parameters tending to one has the same Hausdorff limit $\{0\}$.

At the endpoint, $F_1 = \{\mathrm{id}, x \mapsto x/2\}$ and $F_1(\{0\}) = \{0\}$. This is the smallest invariant set containing Q, so Definition 4.1 gives $A_\bullet = \{0\}$. Definition 4.3 gives the unique upper transition attractor $A^\bullet = \{0\}$.

Let U be any open neighborhood of zero. It contains a nonzero point x, and $K = \{x\} \subset U$ is compact. Formula (2), now with $r_1 = 1$, says

$$h(F_1^n(\{x\}), \{0\}) = |x| > 0 \qquad (n \ge 0).$$

Thus no such U is a basin of attraction for $\{0\}$. This disproves the conjecture's conjunction, even though its uniqueness conclusion holds in this example. □

The source's finite-contractivity-threshold convention also causes no restriction: the same formulas extend to all $t \geq 0$. For $t > 1$, the orbit $g_t^n(x) = r_t^n x$ of any nonzero x is unbounded, ruling out an attractor with an open basin. It also rules out contraction in an equivalent complete metric, since such a contraction would make every g_t orbit converge to its fixed point zero. At $t = 1$ the identity cannot be a contraction in any metric. Both thresholds are therefore one.

Task 2. Characterize singleton attraction for linear IFSs. Express strict attraction to a common fixed point in terms of uniform decay of all matrix products, and identify the role of the joint spectral radius in finite dimensions.

Task 3. Settle the separate uniqueness implication. Determine whether $A_\bullet = A^\bullet$ for one subsequential limit forces all upper transition attractors to agree under (H1)–(H3).

Task 4. Formulate a sufficient terminal-attraction condition. Find a quantitative condition on finite iterates near the common transition set that guarantees an open basin at $t = 1$.

Task 5. Construct examples with larger transition sets. Extend the failure of terminal attraction to families whose coinciding transition attractors have positive dimension, and compare periodic and nonperiodic endpoint isometries.

References

[1] K. Leśniak, N. Snigireva, F. Strobin and A. Vince, *Transition phenomena for the attractor of an iterated function system*, Nonlinearity **35** (2022), 5396–5426. doi:10.1088/1361-6544/ac8af1.